

=====cricket_db=====

-- DROP if exist to allow re-run

DROP TABLE IF EXISTS bowling;

DROP TABLE IF EXISTS batting;

DROP TABLE IF EXISTS players;

-- Players table: player_id is a simple 4-digit integer

```
CREATE TABLE players (  
  player_id INT NOT NULL,          -- 4-digit id (e.g. 1001)  
  full_name VARCHAR(100) NOT NULL,  
  country VARCHAR(50),  
  role ENUM('Batsman','Bowler','All-rounder','Wicket-keeper') DEFAULT 'Batsman',  
  batting_style VARCHAR(30),  
  bowling_style VARCHAR(50),  
  dob DATE,  
  debut_date DATE,  
  active TINYINT(1) DEFAULT 1,  
  jersey_number SMALLINT,  
  PRIMARY KEY (player_id)  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
```

-- Batting stats table: composite PK (player_id, format)

```
CREATE TABLE batting (  
  player_id INT NOT NULL,  
  format ENUM('Test','ODI','T20') NOT NULL DEFAULT 'ODI',  
  matches SMALLINT DEFAULT 0,  
  innings SMALLINT DEFAULT 0,  
  runs INT DEFAULT 0,  
  highest INT DEFAULT 0,  
  batting_avg DECIMAL(5,2) DEFAULT 0.00,  
  strike_rate DECIMAL(6,2) DEFAULT 0.00,  
  hundreds SMALLINT DEFAULT 0,  
  fifties SMALLINT DEFAULT 0,  
  not_outs SMALLINT DEFAULT 0,  
  PRIMARY KEY (player_id, format),  
  CONSTRAINT fk_batting_player FOREIGN KEY (player_id) REFERENCES  
  players(player_id)  
  ON DELETE CASCADE ON UPDATE CASCADE  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
```

-- Bowling stats table: composite PK (player_id, format)

```
CREATE TABLE bowling (  
  player_id INT NOT NULL,  
  format ENUM('Test','ODI','T20') NOT NULL DEFAULT 'ODI',  
  matches SMALLINT DEFAULT 0,  
  innings SMALLINT DEFAULT 0,
```

```

wickets INT DEFAULT 0,
best_figures VARCHAR(10),          -- e.g. '5/27'
bowling_avg DECIMAL(5,2) DEFAULT 0.00,
economy DECIMAL(4,2) DEFAULT 0.00,
strike_rate DECIMAL(6,2) DEFAULT 0.00,
five_wickets SMALLINT DEFAULT 0,
ten_wickets SMALLINT DEFAULT 0,
PRIMARY KEY (player_id, format),
CONSTRAINT fk_bowling_player FOREIGN KEY (player_id) REFERENCES
players(player_id)
ON DELETE CASCADE ON UPDATE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

-- Insert 20 players (IDs 1001..1020)
INSERT INTO players (player_id, full_name, country, role, batting_style, bowling_style, dob,
debut_date, active, jersey_number) VALUES
(1001, 'Rohit Sharma', 'India', 'Batsman', 'Right-hand', 'Right-arm offbreak',
'1987-04-30','2007-06-23', 1, 45),
(1002, 'Virat Kohli', 'India', 'Batsman', 'Right-hand', 'None',
'1988-11-05','2008-08-18', 1, 18),
(1003, 'Jasprit Bumrah', 'India', 'Bowler', 'Right-hand', 'Right-arm fast',
'1993-12-06','2016-01-03', 1, 93),
(1004, 'Joe Root', 'England', 'Batsman', 'Right-hand', 'None',
'1990-12-30','2012-12-10', 1, 66),
(1005, 'Ben Stokes', 'England', 'All-rounder', 'Left-hand', 'Right-arm fast',
'1991-06-04','2011-08-10', 1, 55),
(1006, 'Pat Cummins', 'Australia', 'Bowler', 'Right-hand', 'Right-arm fast',
'1993-05-08','2011-11-17', 1, 30),
(1007, 'David Warner', 'Australia', 'Batsman', 'Left-hand', 'None',
'1986-10-27','2009-01-11', 0, 31),
(1008, 'Kane Williamson', 'New Zealand', 'Batsman', 'Right-hand', 'Off-spin',
'1990-08-08','2010-11-04', 1, 22),
(1009, 'Trent Boult', 'New Zealand', 'Bowler', 'Right-hand', 'Left-arm fast',
'1989-07-22','2011-12-09', 1, 18),
(1010, 'Shaheen Afridi', 'Pakistan', 'Bowler', 'Left-hand', 'Left-arm fast',
'2000-04-06','2018-09-15', 1, 10),
(1011, 'Babar Azam', 'Pakistan', 'Batsman', 'Right-hand', 'None',
'1994-10-15','2015-05-14', 1, 56),
(1012, 'Shakib Al Hasan', 'Bangladesh', 'All-rounder', 'Left-hand', 'Left-arm spin',
'1987-03-24','2006-08-06', 1, 75),
(1013, 'Tamim Iqbal', 'Bangladesh', 'Batsman', 'Left-hand', 'None',
'1989-03-20','2007-02-09', 0, 28),
(1014, 'Kusal Mendis', 'Sri Lanka', 'Batsman', 'Right-hand', 'None',
'1995-02-02','2015-01-03', 1, 20),
(1015, 'Wanindu Hasaranga', 'Sri Lanka', 'All-rounder', 'Right-hand', 'Right-arm
legbreak','1997-02-28','2019-07-18', 1, 49),
(1016, 'Quinton de Kock', 'South Africa', 'Wicket-keeper', 'Left-hand', 'None',
'1992-12-17','2013-12-17', 1, 12),

```

```
(1017, 'Kagiso Rabada', 'South Africa', 'Bowler', 'Right-hand', 'Right-arm fast',
'1995-05-25','2014-11-05', 1, 25),
(1018, 'Jason Holder', 'West Indies', 'All-rounder', 'Right-hand', 'Right-arm fast',
'1991-11-05','2013-02-13', 1, 98),
(1019, 'Nicholas Pooran', 'West Indies', 'Wicket-keeper', 'Left-hand', 'None',
'1995-10-02','2016-01-23', 1, 29),
(1020, 'Andre Russell', 'West Indies', 'All-rounder', 'Right-hand', 'Right-arm fast',
'1988-04-29','2011-06-11', 1, 12);
```

-- Insert ODI batting stats for those 20 players

```
INSERT INTO batting (player_id, format, matches, innings, runs, highest, batting_avg,
strike_rate, hundreds, fifties, not_outs) VALUES
(1001,'ODI',260,243,10000,264,48.50,89.70,29,64,18),
(1002,'ODI',300,287,12000,183,56.32,92.30,43,58,22),
(1003,'ODI',90,82,350,58,15.20,82.40,0,1,4),
(1004,'ODI',160,150,6850,133,50.30,88.40,18,39,12),
(1005,'ODI',194,180,5500,102,42.10,89.20,6,31,9),
(1006,'ODI',77,72,800,38,18.30,75.20,0,0,8),
(1007,'ODI',149,138,8500,179,46.20,95.10,22,46,10),
(1008,'ODI',151,140,7000,133,52.10,85.60,21,34,7),
(1009,'ODI',79,74,500,44,12.20,70.50,0,0,2),
(1010,'ODI',45,40,300,36,13.20,85.00,0,0,3),
(1011,'ODI',125,118,6000,125,56.20,89.90,21,27,6),
(1012,'ODI',245,230,7000,120,40.10,88.00,6,42,11),
(1013,'ODI',150,140,7800,128,35.30,83.70,12,38,5),
(1014,'ODI',136,128,4500,122,34.90,86.40,6,25,8),
(1015,'ODI',54,46,1500,80,32.10,110.40,0,8,3),
(1016,'ODI',140,130,6000,178,45.20,96.20,15,32,9),
(1017,'ODI',120,110,400,48,12.50,78.40,0,0,1),
(1018,'ODI',110,100,2500,95,33.20,84.50,1,15,7),
(1019,'ODI',78,69,2200,75,43.10,110.10,1,18,6),
(1020,'ODI',85,70,2000,88,30.10,140.50,2,8,10);
```

-- Insert ODI bowling stats for those 20 players

```
INSERT INTO bowling (player_id, format, matches, innings, wickets, best_figures,
bowling_avg, economy, strike_rate, five_wickets, ten_wickets) VALUES
(1001,'ODI',260,240,30,'3/22',50.00,5.20,120.00,0,0),
(1002,'ODI',300,285,8,'2/15',62.50,5.10,200.00,0,0),
(1003,'ODI',90,88,200,'5/27',21.10,4.60,29.00,5,0),
(1004,'ODI',160,158,30,'3/20',40.00,4.80,90.00,0,0),
(1005,'ODI',194,180,150,'6/43',29.40,4.90,40.00,3,0),
(1006,'ODI',77,74,250,'7/31',24.80,4.50,22.00,8,0),
(1007,'ODI',149,140,0,NULL,0.00,0.00,0.00,0,0),
(1008,'ODI',151,140,30,'4/22',40.60,4.70,95.00,0,0),
(1009,'ODI',79,78,240,'4/18',25.40,4.60,30.00,0,0),
(1010,'ODI',45,43,180,'6/29',23.80,4.75,29.00,3,0),
(1011,'ODI',125,120,10,'2/21',41.00,4.90,140.00,0,0),
(1012,'ODI',245,230,260,'7/36',29.50,4.80,38.00,7,0),
```

```
(1013,'ODI',150,140,0,NULL,0.00,0.00,0.00,0,0),
(1014,'ODI',136,128,5,'2/18',55.00,5.30,90.00,0,0),
(1015,'ODI',54,50,120,'5/20',22.50,4.20,22.00,4,0),
(1016,'ODI',140,130,0,NULL,0.00,0.00,0.00,0,0),
(1017,'ODI',120,115,210,'6/35',24.00,4.60,25.00,6,0),
(1018,'ODI',110,105,140,'5/29',26.40,4.75,36.00,4,0),
(1019,'ODI',78,74,0,NULL,0.00,0.00,0.00,0,0),
(1020,'ODI',85,70,70,'5/23',25.30,6.20,29.00,2,0);
```

```
=====
```

```
DROP TABLE IF EXISTS emp;
DROP TABLE IF EXISTS dept;
```

```
-- Department table
```

```
CREATE TABLE dept (
  deptno DECIMAL(2,0) NOT NULL,
  dname VARCHAR(14) DEFAULT NULL,
  loc VARCHAR(13) DEFAULT NULL,
  PRIMARY KEY (deptno)
);
```

```
-- Employee table
```

```
CREATE TABLE emp (
  empno DECIMAL(4,0) NOT NULL,
  ename VARCHAR(10) DEFAULT NULL,
  job VARCHAR(9) DEFAULT NULL,
  mgr DECIMAL(4,0) DEFAULT NULL,
  hiredate DATE DEFAULT NULL,
  sal DECIMAL(7,2) DEFAULT NULL,
  comm DECIMAL(7,2) DEFAULT NULL,
  deptno DECIMAL(2,0) DEFAULT NULL,
```

```
PRIMARY KEY (empno),
```

```
-- foreign key linking to dept table
```

```
CONSTRAINT fk_dept FOREIGN KEY (deptno)
  REFERENCES dept(deptno)
  ON DELETE SET NULL
  ON UPDATE CASCADE
```

```
);
```

```
-- Insert departments first
```

```
INSERT INTO dept VALUES ('10','ACCOUNTING','NEW YORK');
INSERT INTO dept VALUES ('20','RESEARCH','DALLAS');
INSERT INTO dept VALUES ('30','SALES','CHICAGO');
```

```

INSERT INTO dept VALUES ('40','OPERATIONS','BOSTON');

-- Insert employees
INSERT INTO emp VALUES
('7369','SMITH','CLERK','7902','1980-12-17','800.00',NULL,'20');
INSERT INTO emp VALUES
('7499','ALLEN','SALESMAN','7698','1981-02-20','1600.00','300.00','30');
INSERT INTO emp VALUES
('7521','WARD','SALESMAN','7698','1981-02-22','1250.00','500.00','30');
INSERT INTO emp VALUES
('7566','JONES','MANAGER','7839','1981-04-02','2975.00',NULL,'20');
INSERT INTO emp VALUES
('7654','MARTIN','SALESMAN','7698','1981-09-28','1250.00','1400.00','30');
INSERT INTO emp VALUES
('7698','BLAKE','MANAGER','7839','1981-05-01','2850.00',NULL,'30');
INSERT INTO emp VALUES
('7782','CLARK','MANAGER','7839','1981-06-09','2450.00',NULL,'10');
INSERT INTO emp VALUES
('7788','SCOTT','ANALYST','7566','1982-12-09','3000.00',NULL,'20');
INSERT INTO emp VALUES
('7839','KING','PRESIDENT',NULL,'1981-11-17','5000.00',NULL,'10');
INSERT INTO emp VALUES
('7844','TURNER','SALESMAN','7698','1981-09-08','1500.00','0.00','30');
INSERT INTO emp VALUES
('7876','ADAMS','CLERK','7788','1983-01-12','1100.00',NULL,'20');
INSERT INTO emp VALUES
('7900','JAMES','CLERK','7698','1981-12-03','950.00',NULL,'30');
INSERT INTO emp VALUES
('7902','FORD','ANALYST','7566','1981-12-03','3000.00',NULL,'20');
INSERT INTO emp VALUES
('7934','MILLER','CLERK','7782','1982-01-23','1300.00',NULL,'10');

```

MySQL Table Creation Scripts

Doctors Table

```

CREATE TABLE Doctors (
    doctor_id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(100),
    specialization VARCHAR(100),
    fees INT

```

```
);
```

Patients Table

```
CREATE TABLE Patients (  
    patient_id INT PRIMARY KEY AUTO_INCREMENT,  
    name VARCHAR(100),  
    age INT,  
    gender VARCHAR(10)  
);
```

Appointments Table

```
CREATE TABLE Appointments (  
    appointment_id INT PRIMARY KEY AUTO_INCREMENT,  
    patient_id INT,  
    doctor_id INT,  
    appointment_date DATE,  
    status VARCHAR(20),      -- Scheduled / Completed / Cancelled  
    FOREIGN KEY (patient_id) REFERENCES Patients(patient_id),  
    FOREIGN KEY (doctor_id) REFERENCES Doctors(doctor_id)  
);
```

Bills Table

```
CREATE TABLE Bills (  
    bill_id INT PRIMARY KEY AUTO_INCREMENT,  
    appointment_id INT,  
    amount DECIMAL(10,2),  
    payment_status VARCHAR(20),  -- Paid / Unpaid  
    FOREIGN KEY (appointment_id) REFERENCES  
Appointments(appointment_id)  
);
```

1. Insert Sample Data — Doctors

```
INSERT INTO Doctors (name, specialization, fees) VALUES
```

```
('Dr. Rajesh Kumar', 'Cardiologist', 1200),  
( 'Dr. Meera Nair', 'Dermatologist', 800),  
( 'Dr. Arjun Singh', 'Orthopedic', 1000),  
( 'Dr. Priya Menon', 'Pediatrician', 700),  
( 'Dr. Sanjay Patel', 'Neurologist', 1500);
```



2. Insert Sample Data — Patients

```
INSERT INTO Patients (name, age, gender) VALUES  
( 'Anandh Kumar', 34, 'Male'),  
( 'Lakshmi Devi', 29, 'Female'),  
( 'John Thomas', 45, 'Male'),  
( 'Sahana R', 12, 'Female'),  
( 'Rohit Sharma', 52, 'Male'),  
( 'Nisha K', 38, 'Female');
```



3. Insert Sample Data — Appointments

```
INSERT INTO Appointments (patient_id, doctor_id, appointment_date,  
status) VALUES  
(1, 1, '2025-01-10', 'Completed'),  
(2, 2, '2025-01-12', 'Completed'),  
(3, 3, '2025-01-15', 'Cancelled'),  
(4, 4, '2025-01-18', 'Completed'),  
(5, 5, '2025-01-20', 'Scheduled'),  
(6, 1, '2025-01-22', 'Completed'),  
(3, 5, '2025-01-25', 'Completed'),  
(2, 1, '2025-02-01', 'Scheduled'),  
(5, 3, '2025-02-03', 'Completed'),  
(1, 2, '2025-02-05', 'Completed');
```



4. Insert Sample Data — Bills

```
INSERT INTO Bills (appointment_id, amount, payment_status) VALUES
(1, 1500.00, 'Paid'),
(2, 800.00, 'Paid'),
(3, 0.00, 'Cancelled'),
(4, 700.00, 'Paid'),
(5, 0.00, 'Unpaid'),
(6, 1200.00, 'Paid'),
(7, 1500.00, 'Unpaid'),
(8, 0.00, 'Unpaid'),
(9, 1000.00, 'Paid'),
(10, 800.00, 'Paid');
```

SQL Queries :

list patients who have visited more than one doctor.

Find appointments where the billed amount is *greater than the doctor's fee*, indicating overcharging.

1.List players with age (calculated from DOB) and their batting average in ODI format

```
Select p.full_name , timestampdiff(year,dob,curdate())
as Age ,b.batting_avg
From players p
Inner join batting b
On p.player_id =b.player_id
Where format='ODI';
```

2.Show full player names in UPPERCASE along with their country and strike rate

```
Select upper(p.full_name),p.country, b.strike_rate
From players p
Inner join batting b
```


On p.player_id =b.player_id

3. List countrywise player count and max runs

Select p.country, count(distinct p.full_name) as
playercount , max(b.runs) as maxruns

From players p

Inner join batting b

On p.player_id =b.player_id

Group by p.country ;

4. List the player whose strike rate is greater than their
country's average strike rate

Use : **players, batting, bowling**

SELECT

p.full_name,

p.country,

b.strike_rate

FROM players p

INNER JOIN batting b

ON p.player_id = b.player_id

WHERE b.strike_rate > (

```
SELECT AVG(b2.strike_rate)  
FROM players p2  
INNER JOIN batting b2  
    ON p2.player_id = b2.player_id  
WHERE p2.country = p.country  
);
```

1.1 INNER JOIN

List all active players along with their ODI batting strike rate.

1.2 LEFT JOIN

Show all players and their Test bowling average. If a player has never bowled in Tests, show the bowling average as NULL.

1.3 RIGHT JOIN

Show all bowling records (any format) and match them with player names, even if the player entry is missing or inactive.

1.4 FULL OUTER JOIN (use UNION as MySQL does not support FULL Join)

Create a list of all batting and bowling formats for each player, showing both values even if one is missing.

1.5 CROSS JOIN

Generate all possible pairs of players with different jersey numbers.

. Get players who have scored more than the avg ODI runs (scalar subquery).

SECTION A — JOIN-BASED QUESTIONS (1–6)

1. Get all players with their ODI batting runs and bowling wickets (INNER JOIN).

Write a query using two joins.

2. List all players showing Test batting stats if available (LEFT JOIN).

Players without Test records must still appear.

3. Show all players who have bowling stats but do not have batting stats (RIGHT JOIN or LEFT ANTI JOIN simulation).

4. Simulate a FULL OUTER JOIN between players and batting using UNION.

5. Display player name, role, batting style, and bowling style with batting ODI strike rate.

Use JOIN + projection.

6. Show the top 10 highest run scorers in ODI with their country (JOIN + ORDER BY + LIMIT).

SECTION B — STRING + DATE FUNCTIONS (7–11)

7. Display players whose names start with 'A' and convert their names to UPPERCASE.

8. Extract birth year of players and list those born before 1990.

Use YEAR(dob).

9. Format debut_date as DD-Month-YYYY and display with full_name.

10. Show bowlers whose best_figures include a 5-wicket haul (string search).

Hint: LIKE '5/%'.

11. Find players whose full_name contains a space and extract the first name using SUBSTRING + LOCATE.

SECTION C — AGGREGATION (12–14)

12. Find total ODI runs made by players from each country.

GROUP BY country.

13. Show players whose total wickets across all formats exceed 100.

Use SUM(wickets) with GROUP BY.

14. List formats and their average strike rates (batting) sorted highest to lowest.

SECTION D — SUBQUERIES (15–18)

15. Find players who scored more than the average ODI runs (scalar subquery).

16. List players who have both batting and bowling records (IN subquery).

17. Get all-rounders using correlated subquery with EXISTS.

18. Display players who have the highest ODI batting average (no LIMIT).

Hint: use = (SELECT MAX())

SECTION E — ADVANCED JOINS + LOGIC (19–20)

19. List players who have Test bowling records but no Test batting records.

Use LEFT JOIN + IS NULL.

20. Display players and the formats where they have both batting and bowling stats.

JOIN + IN (subquery).

SECTION F — TRANSACTIONS, ACID, COMMIT/ROLLBACK (21–23)

21. Start a transaction to update ODI runs for a player and commit it.

Write START TRANSACTION → UPDATE → COMMIT.

22. Update a player's active status inside a transaction and then ROLLBACK.

23. Demonstrate a SAVEPOINT by updating batting first, then bowling, and rolling back to the SAVEPOINT.

SECTION G — DBA (Users, Permissions, Backup) (24–25)

24. Create a new user **cricket_user** with **SELECT + INSERT** permissions only on the cricket database.

25. Write MySQL backup command to export the entire cricket database.

/*SUB-QUERIES*/

/*SINGLE-ROW SQ:*/

```
SELECT * FROM emp WHERE sal > (SELECT sal FROM  
emp WHERE ename = 'JONES');
```

```
SELECT * FROM emp WHERE sal > (SELECT sal FROM  
emp WHERE empno = 7566);
```

/*MULTI-ROW SQ USING: IN / ANY / ALL:*/

**/* > ANY -THE VALUE MUST BE GREATER THAN THE
SMALLEST VALUE IN THE LIST TO TO EVALUATE TO
TRUE*/**

/* < ANY -THE VALUE MUST BE SMALLER THAN THE BIGGEST VALUE IN THE LIST TO TO EVALUATE TO TRUE*/

```
SELECT * FROM emp WHERE sal > ANY (SELECT sal  
FROM emp WHERE sal BETWEEN 2500 AND 4000);
```

```
SELECT * FROM emp WHERE sal < ANY (SELECT sal  
FROM emp WHERE sal BETWEEN 2500 AND 4000);
```

/* > ALL - THE VALUE MUST BE GREATER THAN THE BIGGEST VALUE IN THE LIST TO EVALUATE TO TRUE*/

/* < ALL - THE VALUE MUST BE SMALLER THAN THE SMALLEST VALUE IN THE LIST TO EVALUATE TO TRUE*/

```
SELECT * FROM emp WHERE sal > ALL (SELECT sal FROM emp WHERE sal BETWEEN 2500 AND  
4000);
```

```
SELECT * FROM emp WHERE sal < ALL (SELECT sal FROM emp WHERE sal BETWEEN 2500 AND  
4000);
```

```
SELECT * FROM emp WHERE sal IN (SELECT sal FROM emp WHERE deptno=10);
```

/*NESTED SQ:*/

```
SELECT * FROM emp WHERE sal = (SELECT MAX (sal) FROM  
emp WHERE sal <
```

```
(SELECT MAX (sal) FROM EMP));
```


